Selective Multiband Polarization Conversion with a Compact Asymmetric Reflective Metasurface

***Abstract*—A compact passive reflective metasurface is investigated for selective polarization control across the Ku-, K-, and Ka-band frequency ranges. The unit cell employs an asymmetric metallic strip with a curved section, backed by a continuous ground plane and printed on a 1-mm FR-4 substrate with a periodicity of 1.8 mm. Full-wave simulations produce two cross-polarization-conversion (CPC) regions at 18–26 and 36.5–38.5 GHz, with circular-polarization states near 19.0, 23.5, and 37.5 GHz. The principal CPC response approaches unity, while the circular states are identified using ellipticity and axial ratio in conjunction with the reflection response. The angular study at 0°, 15°, 30°, 45°, and 60° shows that the main resonant conversion remains strong at oblique incidence, although the broad K-band response becomes more selective at larger angles. The combination of multiple conversion characteristics, single-layer construction, compact periodicity, and passive operation makes the proposed surface attractive for frequency-selective polarization control in microwave communication and radar front-end systems.**

***Index Terms*—metasurface, polarization converter, cross-polarization conversion, linear-to-circular conversion, angular stability.**

I. Introduction

Metasurface-based polarization converters have become an important approach for controlling the polarization state of electromagnetic waves because their subwavelength engineered surfaces can provide compact, planar, and multifunctional operation. Recent surveys have highlighted the rapid development of metasurfaces for polarization control, including advances in unit-cell engineering, multifunctionality, angular stability, and emerging intelligent design approaches [1]. A broader review of metasurface polarization devices has likewise established polarization conversion as an important application of metasurfaces, emphasizing their compactness, integration capability, and multifunctional response [2].

Multifunctional reflective metasurfaces have been demonstrated for simultaneous linear-to-linear and linear-to-circular polarization conversion over separated frequency regions [3]. Multi-band reflective structures have further shown efficient linear and circular polarization conversion using compact resonant configurations [4]. Reflection-mode metasurfaces have also been developed to obtain multiband and multifunctional polarization conversion across several frequency regions [5]. Anisotropic metasurfaces have been investigated for ultra-wideband polarization conversion, demonstrating the usefulness of anisotropic resonances for broadband operation [6]. Ultrathin reflective structures have been reported for linear and circular polarization conversion spanning Ku-, K-, and Ka-band frequencies [7]. Ultra-thin converters have additionally combined linear and circular polarization features with RCS-oriented operation and wide-angle performance [8]. Dual-polarization conversion has also been integrated with coding metasurfaces to obtain wideband radar cross-section reduction [9]. Anisotropic reflective converters have demonstrated ultra-broadband linear polarization conversion through multiple resonant responses [10]. Highly miniaturized reflective converters have been proposed to combine linear-to-circular and linear-to-cross-polarization conversion for Ka-band applications with angular-stability analysis [11]. Low-cost single-layer FR-4 metasurfaces have also demonstrated both linear and circular polarization conversion across multiple C-, X-, and Ku-band regions [12].

Conformal polarization conversion has been investigated to extend metasurface functionality from planar to curved platforms, with reflective linear-to-circular conversion demonstrated at millimeter-wave frequencies and angularly stable operation [13]. Multifunctional polarization converters have also been explored for low-RCS radiating systems, demonstrating the integration of polarization manipulation with radar-related applications [14]. Transmission-type dual-band linear-to-circular conversion has been studied for satellite communication, showing the applicability of polarization converters to high-frequency communication links [15]. Low-loss broadband linear-to-circular conversion has further been developed specifically for K-/Ka-band satellite communication applications [16]. Single-layered dual-band converters have also been investigated for linear-to-dual-circular polarization conversion with enhanced angular stability [17]. Ultra-thin reflective metasurfaces with linear and circular polarization characteristics have additionally been developed for Ku-band operation [18].

Despite these developments, simultaneously achieving multiple separated CPC regions and circular-polarization states in a compact reflective structure remains challenging, particularly when high conversion efficiency, single-layer construction, low-cost fabrication, and useful angular stability are required together. This motivates a design that can provide several frequency-selective polarization states without relying on active switching, multilayer alignment, or complex fabrication. Such a configuration is particularly attractive for compact microwave systems operating across the Ku-, K-, and Ka-band frequency ranges.

In this work, a compact asymmetric reflective metasurface is investigated for selective linear-to-cross-linear and linear-to-circular polarization conversion across the Ku-, K-, and Ka-bands. The proposed single-layer structure employs an

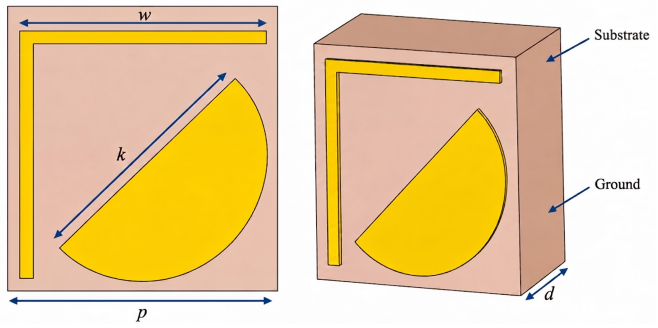


Fig. 1. Front and perspective views of the proposed reflective unit cell.

TABLE I
OPTIMIZED GEOMETRICAL PARAMETERS OF THE PROPOSED UNIT CELL.

Parameter	Description	Value (mm)
x	Strip width	0.10
w	Strip length	1.65
t	Conductor thickness	0.035
p	Unit-cell period	1.80
k	Geometric length	1.70
d	Substrate thickness	1.00

asymmetric metallic strip and curved section on a 1-mm FR-4 substrate with a periodicity of 1.8 mm. It provides two CPC regions at 18–26 and 36.5–38.5 GHz, with circular-polarization states near 19.0, 23.5, and 37.5 GHz. The polarization response is evaluated using reflection coefficients, polarization conversion ratio, ellipticity, and axial ratio, while the angular performance is investigated up to 60° incidence.

II. Unit-Cell Design and Analysis Method

A. Geometry and Materials

The proposed unit cell consists of an asymmetric metallic strip connected to a curved section. The patterned conductor is placed on a 1-mm FR-4 dielectric layer and backed by a continuous metallic ground plane. The unit-cell period is $p = 1.8$ mm, and the conductor thickness is $t = 0.035$ mm. The remaining optimized dimensions are listed in Table I. The absence of rotational symmetry produces different electromagnetic responses along the two orthogonal axes, providing the basis for polarization conversion.

B. Reflection Model

For a linearly polarized incident wave, the reflected field is represented by the Jones matrix

$$\mathbf{R} = \begin{bmatrix} R_{xx} & R_{xy} \\ R_{yx} & R_{yy} \end{bmatrix}. \tag{1}$$

For y-polarized incidence, the PCR is evaluated from the cross- and co-polarized reflection coefficients as

$$\mathrm{PCR}_y = \frac{|R_{xy}|^2}{|R_{xy}|^2 + |R_{yy}|^2}. \tag{2}$$

The same definition is applied to x-polarized incidence by interchanging the polarization indices. A PCR close to one means that most of the reflected power is transferred into the orthogonal linear component. Circular polarization is determined from the relative amplitude and phase of the two orthogonal reflected fields. Ellipticity is used to identify the handedness, while AR provides an additional measure of how close the reflected state is to circular polarization.

C. Simulation Procedure

The structure is analyzed in CST Microwave Studio using periodic boundary conditions in the transverse directions and Floquet-port excitation along the propagation direction. The frequency sweep covers 5–40 GHz. Both x- and y-polarized normal incidence are considered. For angular analysis, the incidence angle is varied from 0° to 60°. The optimization targets strong cross-polarized reflection at the principal resonances, useful circular states between the resonances, compact geometry, and retention of the main conversion features under oblique incidence.

III. Results and Discussion

A. Reflection Characteristics

Fig. 2 shows the simulated reflection coefficients in linear scale. The co-polarized terms exhibit pronounced minima around the main resonances, while the corresponding cross-polarized terms rise strongly. The dominant resonant features occur near 19, 23.5, and 37 GHz. This complementary behavior indicates that the incident linear field is being predominantly redirected to the orthogonal reflected component at the resonant frequencies.

The dB-scale response in Fig. 3 gives a clearer view of the suppression of the co-polarized field. Around 19 GHz, the co-polarized level reaches approximately -29 dB, and a deeper minimum of about -35 dB appears near 23.5 GHz. A further pronounced dip is observed around the 37-GHz resonance. These minima coincide with high cross-polarized reflection and form the basis of the broad K-band and upper Ka-band CPC responses.

B. CPC and Circular-Polarization Responses

The PCR response in Fig. 4 identifies two main CPC regions. The first extends from 18 to 26 GHz, where PCR remains close to unity over a comparatively broad interval. The second occurs from 36.5 to 38.5 GHz, with a peak of approximately 0.95 near 37.5 GHz. The separated response is useful because the surface can provide a high cross-linear state without changing its physical configuration.

The ellipticity curve in Fig. 5 provides complementary information on the reflected polarization states and their handedness. Circular-polarization states are identified near 19.0, 23.5, and 37.5 GHz based primarily on the pronounced AR minima at these frequencies. The ellipticity response provides additional information on the polarization handedness, while AR provides a direct measure of the proximity to circular polarization.

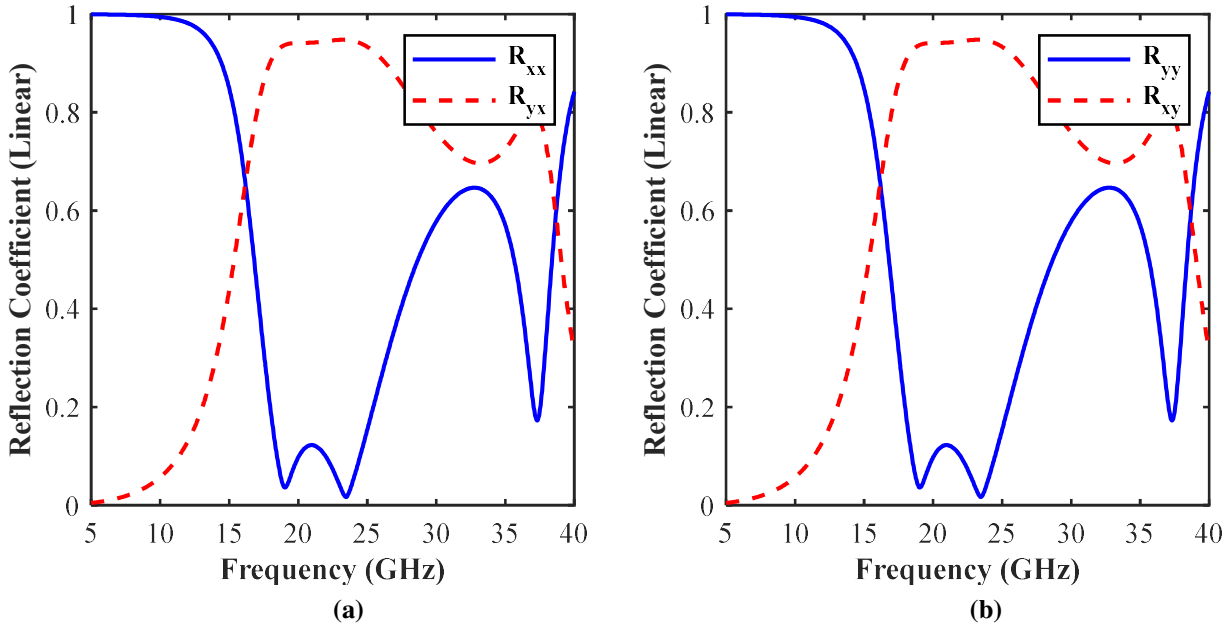


Fig. 2. Linear-scale co- and cross-polarized reflection coefficients for (a) x-polarized and (b) y-polarized incidence.

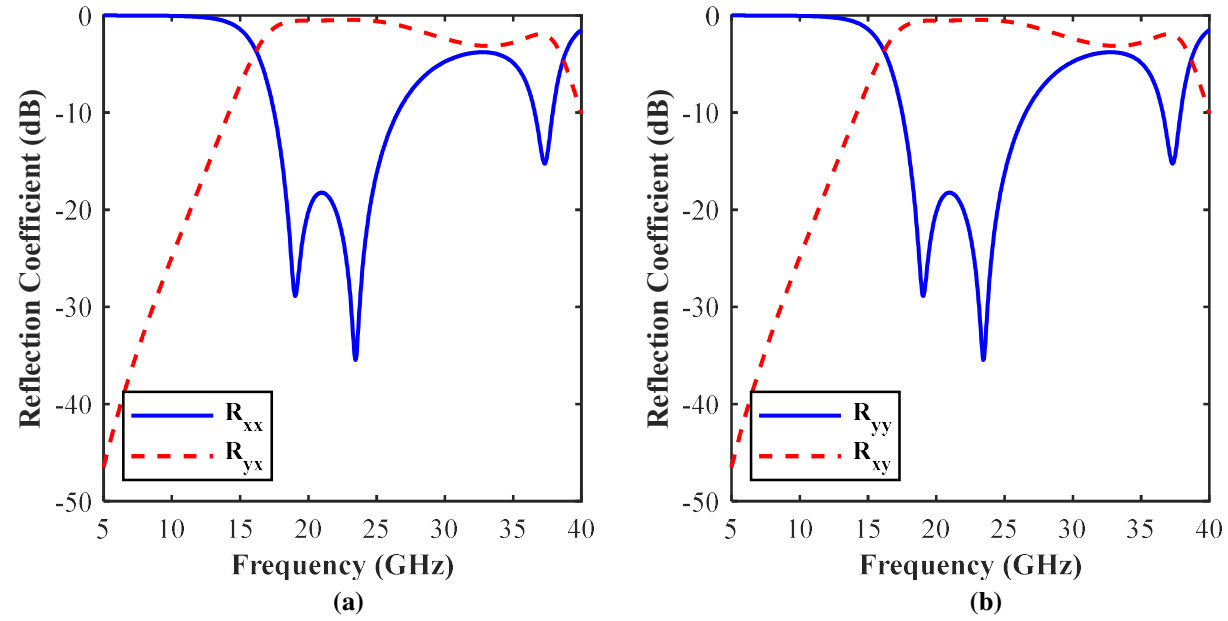


Fig. 3. dB-scale co- and cross-polarized reflection coefficients for (a) x-polarized and (b) y-polarized incidence.

TABLE II
PRINCIPAL POLARIZATION-CONVERSION CHARACTERISTICS.

Mode	Frequency (GHz)	State
CPC-1	18.0–26.0	Cross-linear
CP-1	∼19.0	Circular
CP-2	∼23.5	Circular
CPC-2	36.5–38.5	Cross-linear
CP-3	∼37.5	Circular

C. Axial-Ratio Verification

The AR response in Fig. 6 further verifies the circular-polarization states. Pronounced minima occur near 19.0, 23.5, and 37.5 GHz, where the AR approaches 0 dB, indicating nearly equal orthogonal reflected components with a suitable phase relationship. The AR remains relatively low between 19.0 and 23.5 GHz, with a moderate hump near 21 GHz. In contrast, high AR peaks near 16.2, 33.0, and 39.0 GHz indicate highly linear-polarized reflected fields. Thus, ellipticity

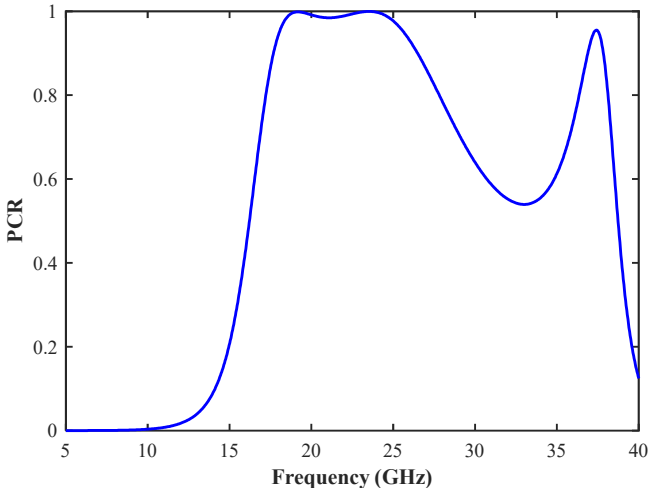


Fig. 4. Polarization conversion ratio under normal incidence.

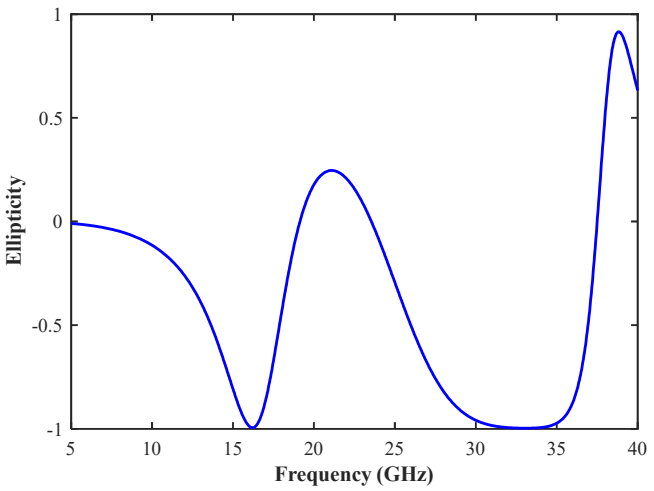


Fig. 5. Ellipticity of the reflected field under normal incidence.

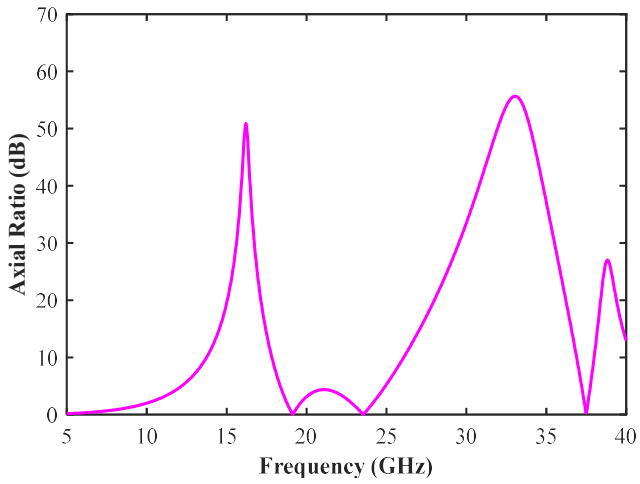


Fig. 6. Axial ratio of the reflected wave over 5–40 GHz.

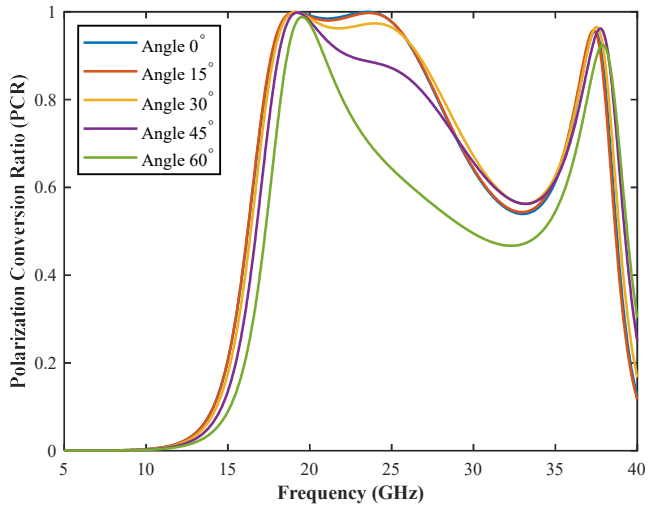


Fig. 7. PCR response for incidence angles of 0°, 15°, 30°, 45°, and 60°.

and AR together provide complementary identification of the reflected polarization states, while PCR quantifies orthogonal power conversion.

D. Angular Stability

The angular response is evaluated at 0°, 15°, 30°, 45°, and 60°, as shown in Fig. 7. Up to 30°, the principal PCR features remain close to their normal-incidence behavior with only small spectral displacement. At 45°, the broad K-band plateau becomes lower, reaching roughly 0.85–0.90 over part of the band. At 60°, the response becomes more selective, but the principal resonances remain pronounced, including about 0.98 near 19.5 GHz and above 0.90 near 37.5 GHz. These results show that the proposed structure maintains strong conversion at its main resonant frequencies even under oblique incidence.

Thus, the 60° result is best understood as strong resonant angular stability rather than complete preservation of the full broadband plateau. This distinction is relevant when comparing the design with converters that report wide-angle operation using a fixed PCR threshold.

IV. Comparative Assessment

Table III compares the proposed design with representative converters. The comparison focuses on substrate, CPC and circular-polarization characteristics, and angular stability. The compared designs cover multifunctional, multiband, ultrathin, low-cost, and angular-stable polarization conversion. The proposed design combines two CPC regions and circular-polarization states in a passive single-layer structure while retaining strong principal resonant conversion at large incidence angles.

The proposed structure generates all conversion states from a fixed metallic geometry; therefore, no bias network or switching element is required. This is an important practical distinction from active reconfigurable approaches and from multilayer implementations that require additional alignment

TABLE III
COMPARISON OF THE PROPOSED DESIGN WITH REPRESENTATIVE POLARIZATION CONVERTERS.

Ref.	Substrate	CPC Band (GHz)	LTC/CP Characteristic (GHz)	Angular Stability
[3]	FR-4 (1.6 mm)	5.6–6.5, 9.6–11.9	5.4, 6.9–8.9	30°
[4]	Rogers RO3003 (1.52 mm)	9.12–9.57, 13.08–14.07	8.37–8.97, 14.50–18.66	45°
[5]	PTFE (1.6 mm)	13.10–14.72, 25.43–26.00	16.08–24.16, 27.82–32.21	30°
[7]	Rogers RT5880 (1.57 mm)	15.97–16.85, 24.70–26.65	17.27–23.92, 27.47–33.85	45°
[8]	Taconic TLY-5 (0.8 mm)	18.55–26.90, 43.55–45.30	31.8–40.5	40°
[9]	FR-4 (1.0 mm)	12.37–13.03, 18.96–32.37	13.17–18.44, 33.33–40.35	40°
[11]	FR-4 (1.0 mm)	11.41–12.00, 19.01–22.34, 40.74–46.82	10.60–10.92, 12.12–17.32, 22.72–37.76	45°
[12]	FR-4 (1.6 mm)	6.46–6.78, 10.52–11.85, 16.49–17.37	7.28–9.40, 13.38–15.19	45°
This work	**FR-4 (1.0 mm)**	**18.0–26.0, 36.5–38.5**	**∼19.0, ∼23.5, ∼37.5**	**60°**

and fabrication steps. The frequency-selective response provides circular-polarization states near 19.0, 23.5, and 37.5 GHz within the principal CPC regions.

The comparison in Table III shows that the proposed design provides a balance between functionality and structural simplicity. The compared designs demonstrate multifunctional or multiband conversion, broadband anisotropic operation, thin reflective configurations, RCS-oriented conversion, and angularly stable polarization control. In contrast, the proposed design obtains two CPC regions and circular-polarization states from a passive asymmetric single-layer cell and retains strong principal resonances up to 60°.

The present results are numerical. Finite-array effects, fabrication tolerance, dielectric-property variation, conductor roughness, and measurement uncertainty may alter the exact resonance locations or conversion levels. In particular, the 36.5–40 GHz transition region should be verified experimentally. Therefore, the reported 60° angular result is interpreted as retention of strong principal resonant conversion rather than preservation of the complete broadband response at a fixed PCR threshold.

V. Conclusion

A compact passive reflective metasurface is investigated for selective polarization control across the Ku-, K-, and Ka-band frequency ranges. The proposed 1.8-mm-period unit cell on 1-mm FR-4 provides two CPC regions at 18–26 and 36.5–38.5 GHz, with circular-polarization states near 19.0, 23.5, and 37.5 GHz. The polarization states are identified from the combined reflection, PCR, ellipticity, and AR responses. Angular analysis demonstrates strong resonant conversion up to 60°, although the broadband response becomes more selective at larger incidence angles. The proposed single-layer structure therefore provides a compact and passive approach to multifunctional, frequency-selective polarization control. Future work will focus on experimental validation, fabrication tolerance analysis, and further improvement of angular stability for practical K-/Ka-band applications.

References

[1] H. Z. Khan, J. Zafar, A. Jabbar, J. ur R. Kazim, M. Ur Rehman, M. A. Imran, and Q. H. Abbasi, "Advancements in metasurfaces for polarization control: A comprehensive survey," *Next Res.*, vol. 2, no. 3, p. 100407, 2025.

[2] Z. Shen and X. Lin, "A review of metasurface polarization devices," *Opt. Mater.*, vol. 146, p. 114567, 2023.

[3] M. A. Shah *et al.*, "Multifunctional metasurface for linear-to-linear and linear-to-circular polarization conversion," *IEEE Access*, vol. 14, pp. 18169–18180, 2026.

[4] J. Zafar *et al.*, "Multi-band reflective metasurface for efficient linear and circular polarization conversion," *Opt. Quantum Electron.*, vol. 57, no. 2, 2025.

[5] S. Hafeez, J. Yu, F. A. Umrani, W. Yun, and M. Ishfaq, "A multiband and multifunctional metasurface for linear and circular polarization conversion in reflection modes," *Crystals*, vol. 14, no. 3, p. 266, 2024.

[6] F. D. M. Nobre, M. W. B. da Silva, A. L. P. de Siqueira Campos, and T. N. Ferreira, "Analysis and design of an ultra-wideband polarization converter based on anisotropic metasurface," in *Proc. 20th SBMO/IEEE MTT-S Int. Microw. Optoelectron. Conf. (IMOC)*, 2023, pp. 151–153.

[7] H. Z. Khan, A. Jabbar, J. ur R. Kazim, M. Ur Rehman, M. A. Imran, and Q. H. Abbasi, "Multi-band ultrathin reflective metasurface for linear and circular polarization conversion in Ku, K, and Ka bands," *Commun. Eng.*, vol. 3, no. 1, 2024.

[8] F. Alahmed Albasry, G. Ozturk, U. C. Hasar, and M. F. Corapsiz, "Ultra-thin metasurface polarization converter with linear and circular polarization features for RCS applications," *Arab. J. Sci. Eng.*, vol. 50, no. 8, pp. 5463–5473, 2025.

[9] S. Hafeez, J. Yu, F. A. Umrani, Y. Huang, W. Yun, and M. Ishfaq, "Dual-polarization conversion and coding metasurface for wideband radar cross-section reduction," *Photonics*, vol. 11, no. 5, p. 454, 2024.

[10] J. Xu, R. Li, S. Wang, and T. Han, "Ultra-broadband linear polarization converter based on anisotropic metasurface," *Opt. Express*, vol. 26, no. 20, pp. 26235–26241, 2018.

[11] M. A. Shukoor and S. Dey, "Highly miniaturized wideband angular stable linear-circular and linear-cross polarization converter for Ka-band satellite applications," *Int. J. Microw. Wireless Technol.*, vol. 15, no. 9, pp. 1557–1569, 2023.

[12] Y. Kaya, "Cheaper, wide-band, ultra-thin, and multi-purpose single-layer metasurface polarization converter design for C-, X-, and Ku-band applications," *Symmetry*, vol. 15, no. 2, p. 442, 2023.

[13] V. S. Yadav, D. Kundu, B. K. Kaushik, and A. Patnaik, "Metasurface-based conformal linear-to-circular polarization converter for emerging wireless applications," Optik, vol. 296, p. 171553, 2024.

[14] M. A. Sofi, K. Saurav, S. K. Koul, and E. L. Tan, "Wideband multi-functional polarization converter: Application to low radar cross-section radiating system," *IETE J. Res.*, vol. 70, no. 6, pp. 5606–5616, 2024.

[15] P. Naseri, S. A. Matos, J. R. Costa, C. A. Fernandes, and N. J. G. Fonseca, "Dual-band dual-linear-to-circular polarization converter in transmission mode: Application to K/Ka-band satellite communications," IEEE Trans. Antennas Propag., vol. 66, no. 12, pp. 7128–7137, 2018.

[16] Y. Geng, Y. Cong, K. Gong, and Y. Zhang, "A broadband and low-loss linear-to-circular polarization converter for K-/Ka-band satellite communications," *IEEE Antennas Wireless Propag. Lett.*, vol. 24, no. 10, pp. 3714–3718, 2025.

[17] Y. Chen, G. Zhai, S. Wang, J. Gao, and J. Ren, "Dual-band single-layered linear-to-dual-circular polarization converter with high angle stability for satellite communication," *IEEE Antennas Wireless Propag. Lett.*, vol. 22, no. 8, pp. 2017–2021, 2023.

[18] G. Ozturk and M. F. Corapsiz, "Ultra-thin reflective linear and circular polarization converter for Ku band applications," *Opt. Commun.*, vol. 516, p. 128268, 2022.